

SIE 433/533 Fundamentals of Data Science for Engineer

Homework 5

Due: April 24 (11:59PM), 2025

Instruction:

There are **4 (four) problems** in total. Each of them may contain several sub-problems.

If you want to copy&paste your code and highlight the code, (1) go to <https://highlight.hohli.com/> (2) choose Python as the language (3) copy your code to the console, and (4) click *Highlight*.

Submission Guidelines:

- Ensure to compile all your solutions into a **single PDF file** and submit it via D2L before the deadline.
- **Detailed intermediate steps are required for scoring.** Solutions that present only the final answer without supporting work will not be awarded any points.
- For problems solved using Python, **include the corresponding code within your PDF submission.** To aid in evaluating your work, please highlight the key segments of your code. This will streamline the review process, particularly in identifying any errors in your solution approach.

Problem 1: Linear SVM-based three-class classification for Iris dataset

In the script provided for the class i.e. “*SVM.ipynb*”, we only apply the Iris dataset for binary classification. In fact, there are 3 classes of targets in the Iris dataset. Hence, in this problem, we are going to have similar practices but for a three-class classification. Hard margin and soft margin-based linear SVM are required to apply here. In this problem,

- a. Please train three different support vector machine models with different criterion
 - 1) Hard-margin, with $C = 1e-5$
 - 2) Soft-margin, with $C = 1$
 - 3) Soft-margin, with $C = 800$
- b. Report their accuracies by using *accuracy_score* from *sklearn.metrics*. By accuracies, you do not need to use *train_test_split* to split the data into two sets, but simply test the performance of the model in the whole dataset

Problem 2: Nonlinear SVM-based three-class classification for Iris dataset

You are required to apply the RBF and polynomial kernel-based SVM to have a three-class classification on the Iris dataset. Perform the hard-margin and soft-margin on each kernel method.

- a. Please train four different models
 - 1) RBF kernel, $\gamma=0.7$, $C = 1e-5$
 - 2) RBF kernel, $\gamma=0.7$, $C = 1$
 - 3) Polynomial kernel, $\text{degree} = 4$, $C = 1e-5$
 - 4) Polynomial kernel, $\text{degree} = 4$, $C = 1$
- b. Report their accuracies by using *accuracy_score* from *sklearn.metrics*. By accuracies, you do not need to use *train_test_split* to split the data into two sets, but simply test the performance of the model in the whole dataset

Problem 3. Compare the accuracies calculated by the models from problem 1 and problem 2. Which model is the best? Explain. (Please write the answer in the Jupyter Notebook as a markdown)

Problem 4: KNN-based three-class classification for the Iris dataset

Implement KNN algorithm for different numbers of neighbors (10, 20, 30), set $p=1$, `metric="minkowski"`

- a. Fill the number of neighbors for the *for* loop
- b. Create an instance of the KNN classifier and fit the data.
- c. Fit the model
- d. predict the value for *xx* and *yy* in the mesh, and assign the predicted value to *Z*.